

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

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Class :SYIT	Batch : 1
Date of Assignment :12-08-26	Date/Time of Submission :12-08-26

Part – A Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

CODE:-

```
import pandas as pd

data = {
    "Name": ["Dhruv", "Niraj", "Sumeet"],
    "Age": [20, 21, 19],
    "Marks": [85, 90, 88]
}

df = pd.DataFrame(data)
print(df)
print("\nSumeet s019")
```

OUTPUT:-

```
===== RESTART: C:/Usr
      Name  Age  Marks
0   Dhruv   20     85
1   Niraj   21     90
2  Sumeet   19     88

Sumeet s019
```

5b. Write a python code to show statistical information on given data set.

CODE:-

```
import statistics

print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.mode([1, 1, 3, -5, 7, -9, 11]))
print(statistics.stdev([1, 30, 50, 100]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print("\nSumeet s019")
```

OUTPUT:-

```
===== RESTART
7
7
1
41.67633221226007
70.80333333333333

Sumeet s019
```

5c. Write a python code to create pandas series from dictionaries.

CODE:-

```
import pandas as pd
dictionary = {'A': 50, 'B': 10, 'C': 80}
series = pd.Series(dictionary, index=['B', 'C', 'A'])

print(series)
print("\nSumeet S019")
```

OUTPUT:-

```
===== RESTART
B      10
C      80
A      50
dtype: int64

Sumeet S019
```

CODE:-

```
import pandas as pd
dictionary = {'NIRAJ': 10, 'RUDRA': 20, 'DHRUV': 30}
series = pd.Series(dictionary)

print(series)
print("\nSumeet S019")
```

OUTPUT:-

```
===== RES'
NIRAJ      10
RUDRA      20
DHRUV      30
dtype: int64

Sumeet S019
>
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

CODE:-

```
import pandas as pd
dict = {'name': ["Dhruv", "Aryan", "Niraj", "Sumeet"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score': [90, 40, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, False, True])

print(df)
print("\nSumeet S019")
```

OUTPUT:-

```
===== RESTART: C:/Users/mvl
      name  degree  score
True  Dhruv    MBA     90
False  Aryan    BCA     40
False  Niraj  M.Tech     80
True   Sumeet    MBA     98

Sumeet S019
.
```